Day: Date:

Assignment # 3:

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SP25-BSE-082

F.DLD

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1) Flip-flop equations:-

F-F	J	K
A(top)	$J\_A = XB$	$K\_A = X'$
B(bottom)	$J\_B = X + A$	$K\_B = X'$

2) State Equations:

using JK characteristic equation:

$Q(t+1) = J \cdot Q' + K' \cdot Q$

For F.F A:

$A(t+1) = J\_A \cdot A' + K'\_A \cdot A =$

$XB \cdot A' = X \cdot A = X(A+B)$

For F-F B:

$B(t+1) = J\_B \cdot B' + K'\_B \cdot B = (X+A) \cdot B'$

$+ X \cdot B = X + AB'$

3) Output Equation:

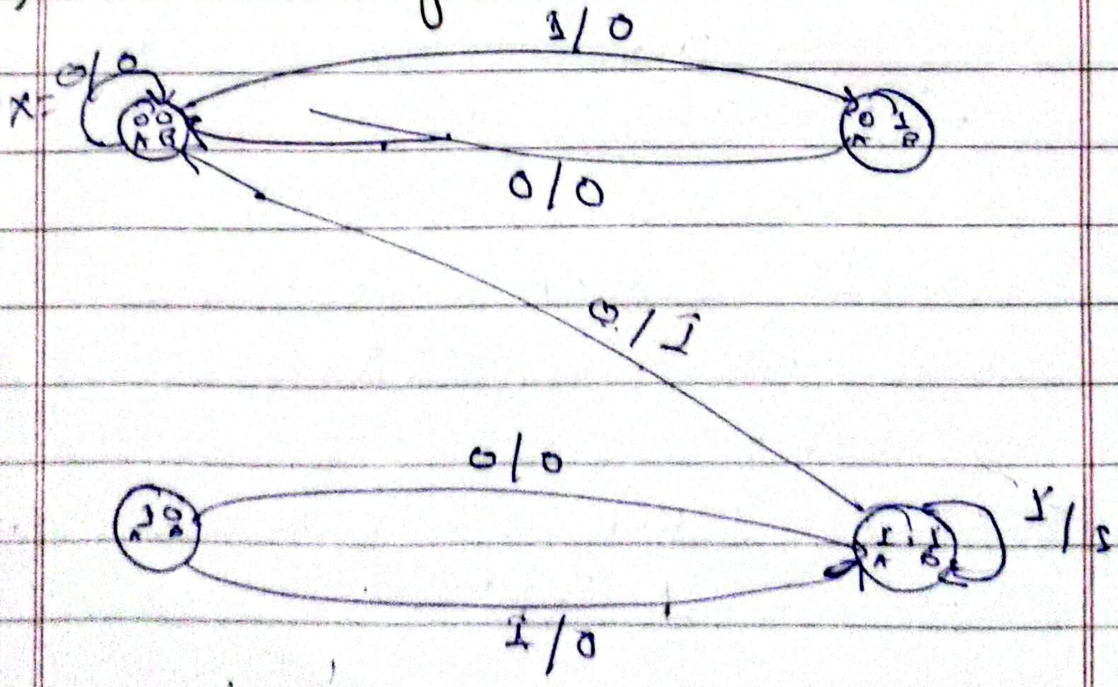
$Y = A \cdot B = Q_1 \cdot Q_2$

4) State Table

Date:

Present State (A,B)	Input X	A^+, B^+	Y
0,0	0	0,0	0
0,0	1	0,1	0
0,1	0	0,0	0
0,1	1	1,1	0
1,0	0	0,1	0
1,0	1	1,1	0
1,1	0	0,0	1
1,1	1	1,1	1

5) state Diagram:



Y=1 transitions (A=B=1)